Author: ML Engineering Lead | Target: Production Architectures | Status: Validation Complete

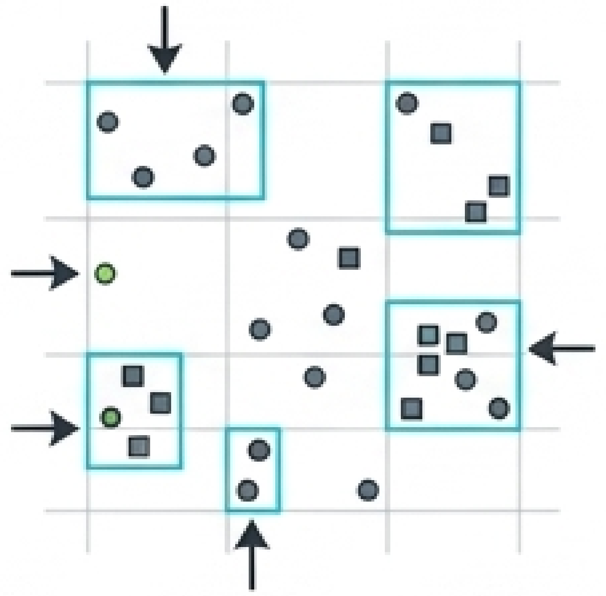
Advanced Segmentation Architectures: Resolving the Annotation Paradox

Weakly Supervised Frameworks, Custom Loss Formulations, and Foundation Models for High-Precision Spatial Mapping

Strategic Objective: Bridging the Supervision Gap

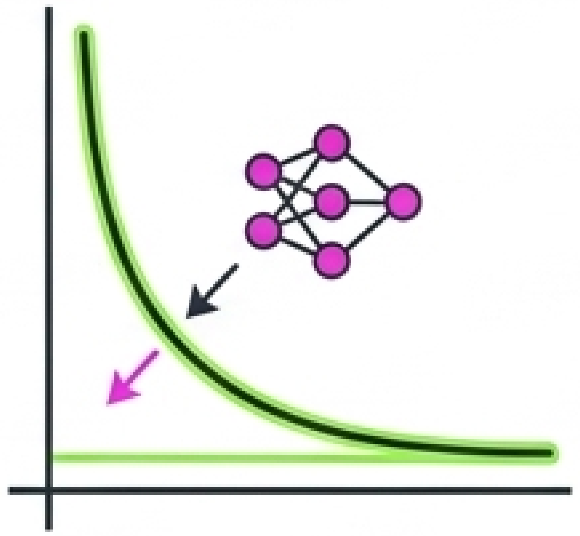
Deploying weakly supervised frameworks to map sparse input data to dense pixel-level classifications.

Sparse Data Ingestion



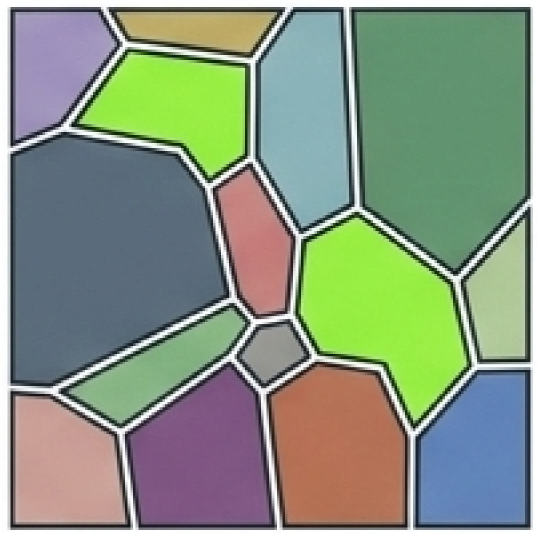
Utilize **points** & **bounding boxes** to cut overhead.

Algorithmic Adaptation



Custom **pCE Loss** & Distilled SAM Prompters.

Dense Prediction



Pixel-perfect segmentation output.

The Annotation Paradox

Reduces labeling overhead by up to 90% — but breaks standard Cross-Entropy loss.

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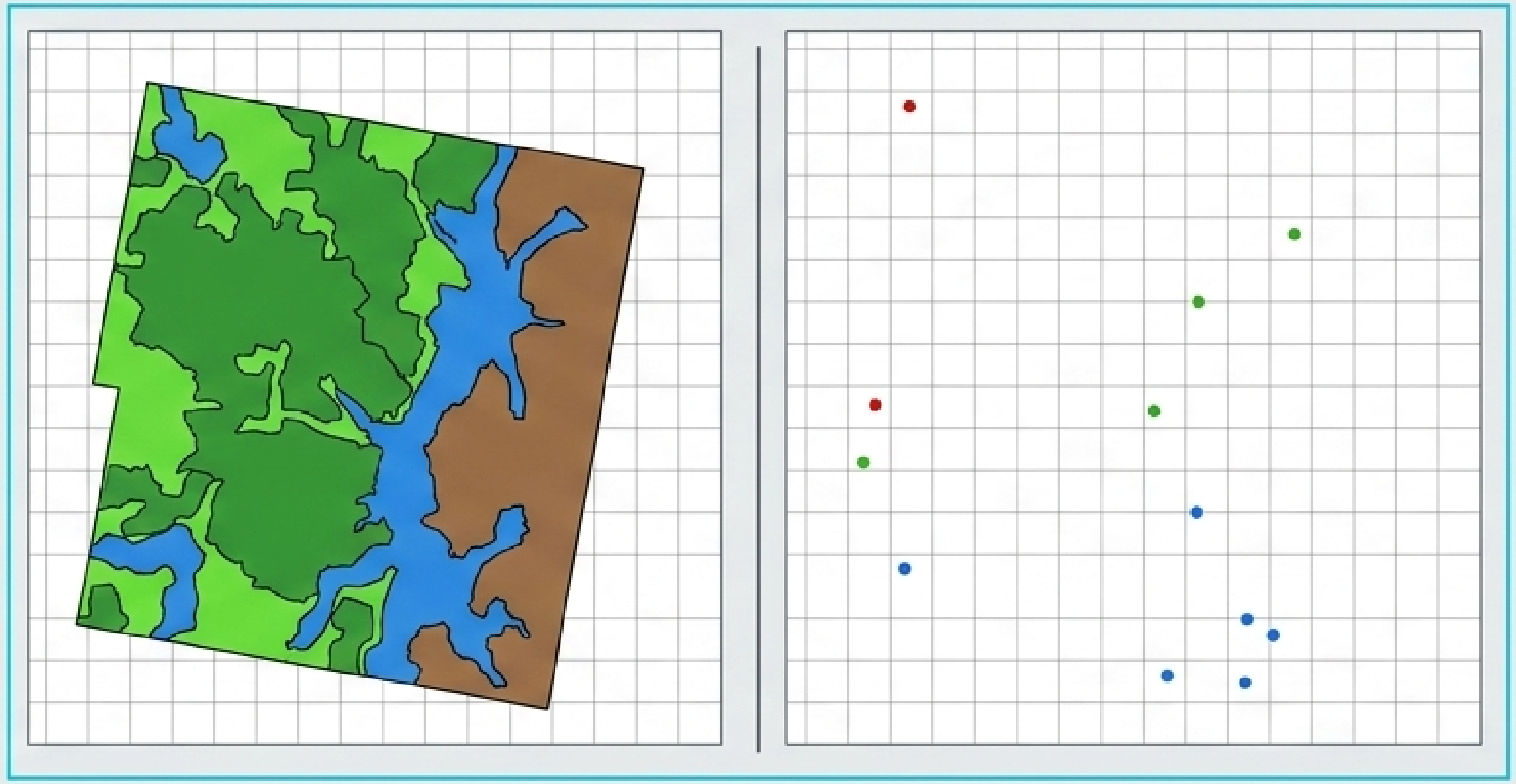
WARNING: Highly
challenging non-
standard optimization
problem.
```



Mask - Complete Supervision | Points - Very Limited Info

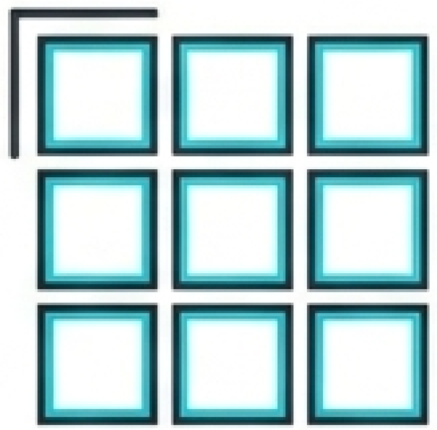
Baseline Preprocessing Bottlenecks (U-Net)

Dataset Context: Dubai Aerial Dataset (72 source tiles -> 1,305 operational patches).

Dimensional Cropping

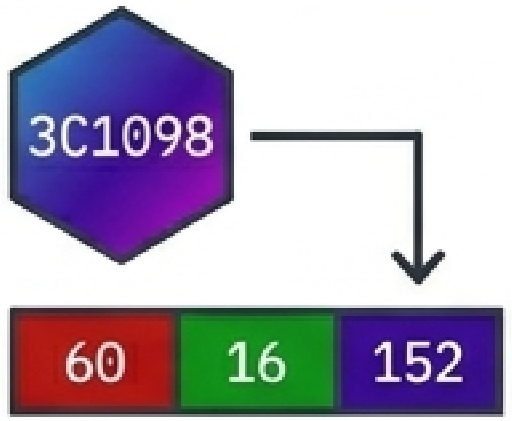
Target nearest divisible by 256. Avoid resize/interpolation artifacts.

Patchification



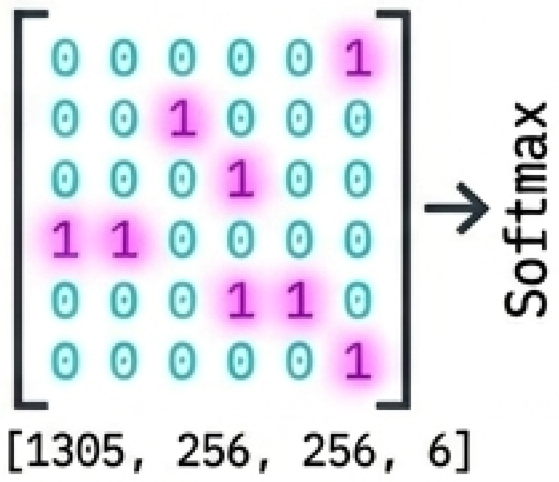
Extract [256, 256, 3] tensor patches.

Hex-to-RGB Mapping



3C1098 -> RGB(60, 16, 152)

Categorical Encoding



Integer mapping -> One-Hot [1305, 256, 256, 6] for Softmax.

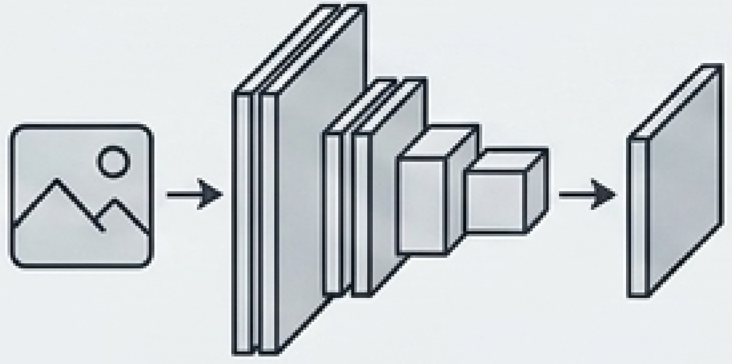
The Supervision Spectrum

Dimension	Fully Supervised (U-Net)	Weakly Supervised RS (Landvisor)	Foundation-Assisted (WPS-SAM)
Label Density	Complete Masks	Sparse Points	Bounding Boxes
Loss Function	Standard CE / Dice	Partial CE (pCE)	Cross-Entropy via Prompter
Backbone Compute	VGG / ResNet	Pre-trained ImageNet	Frozen SAM-ViT-B
Operational Overhead	Max	Low	Low

Landvisor RS: Weakly Supervised Framework

Transfer Learning

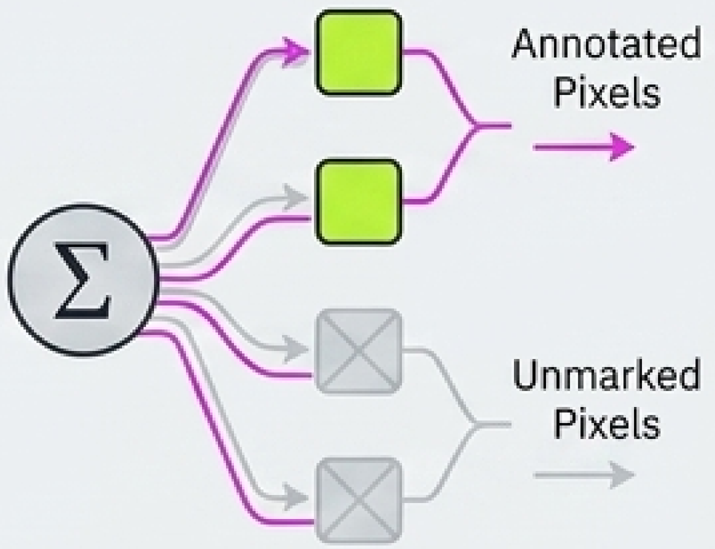
`ImageNet` backbones for feature extraction.



Partial CE Loss

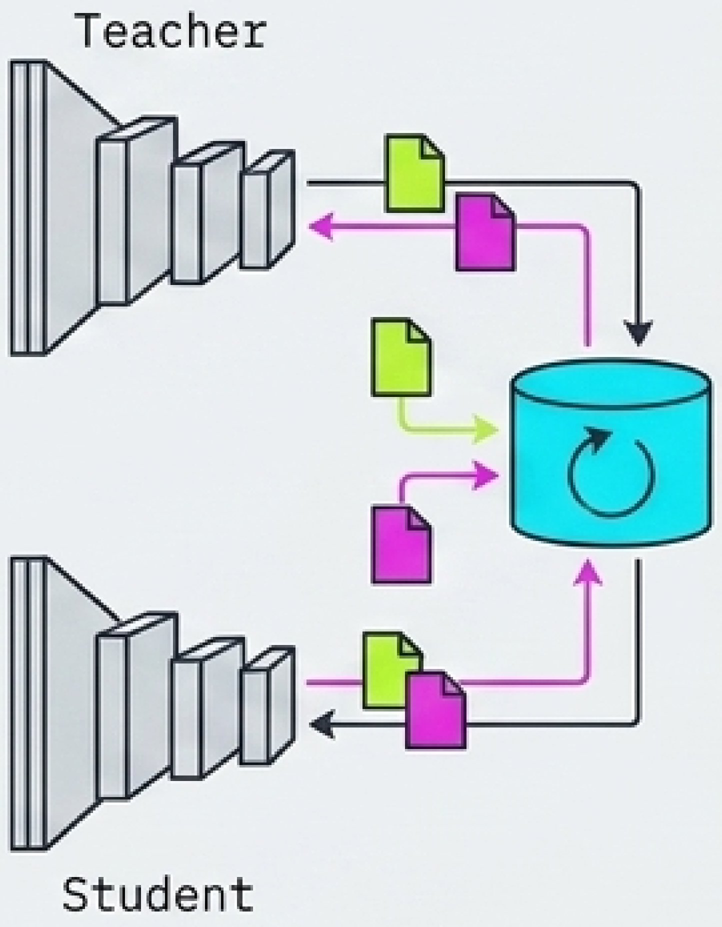
Custom gradient routing.

$$\mathrm{pfCE} = \frac{\sum(\text{Focal loss}(\text{pre}, \text{GT}) \times \text{MASK}_{\text{labeled}})}{\sum \text{MASK}_{\text{labeled}}}$$



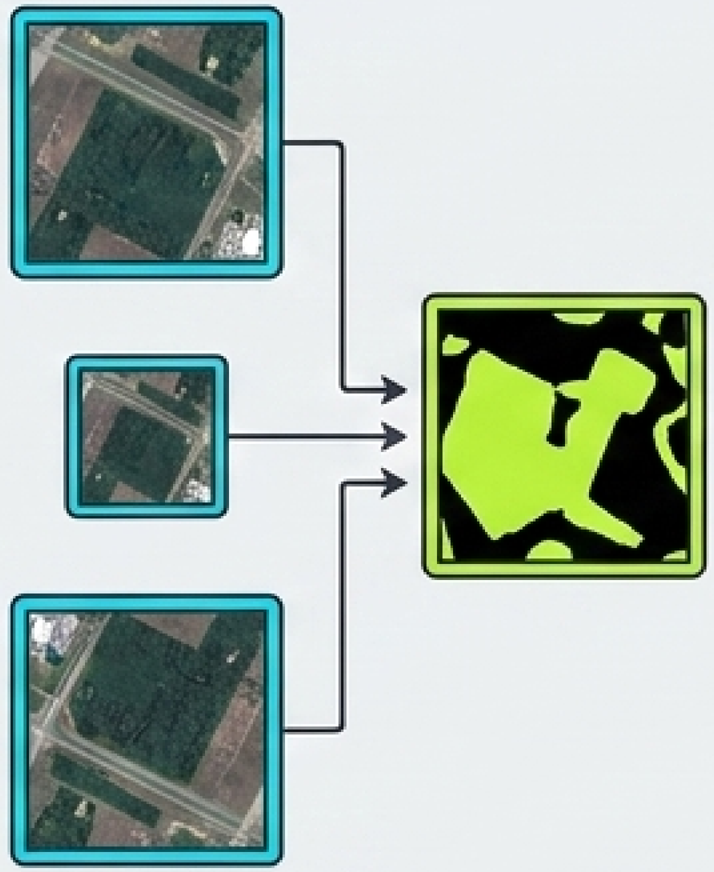
Semi-Supervised Bank

Teacher/Student network updating reliable vs. unreliable pixels.



Ensemble & TTA

Test-Time Augmentation via multi-scale flipping for variance reduction.



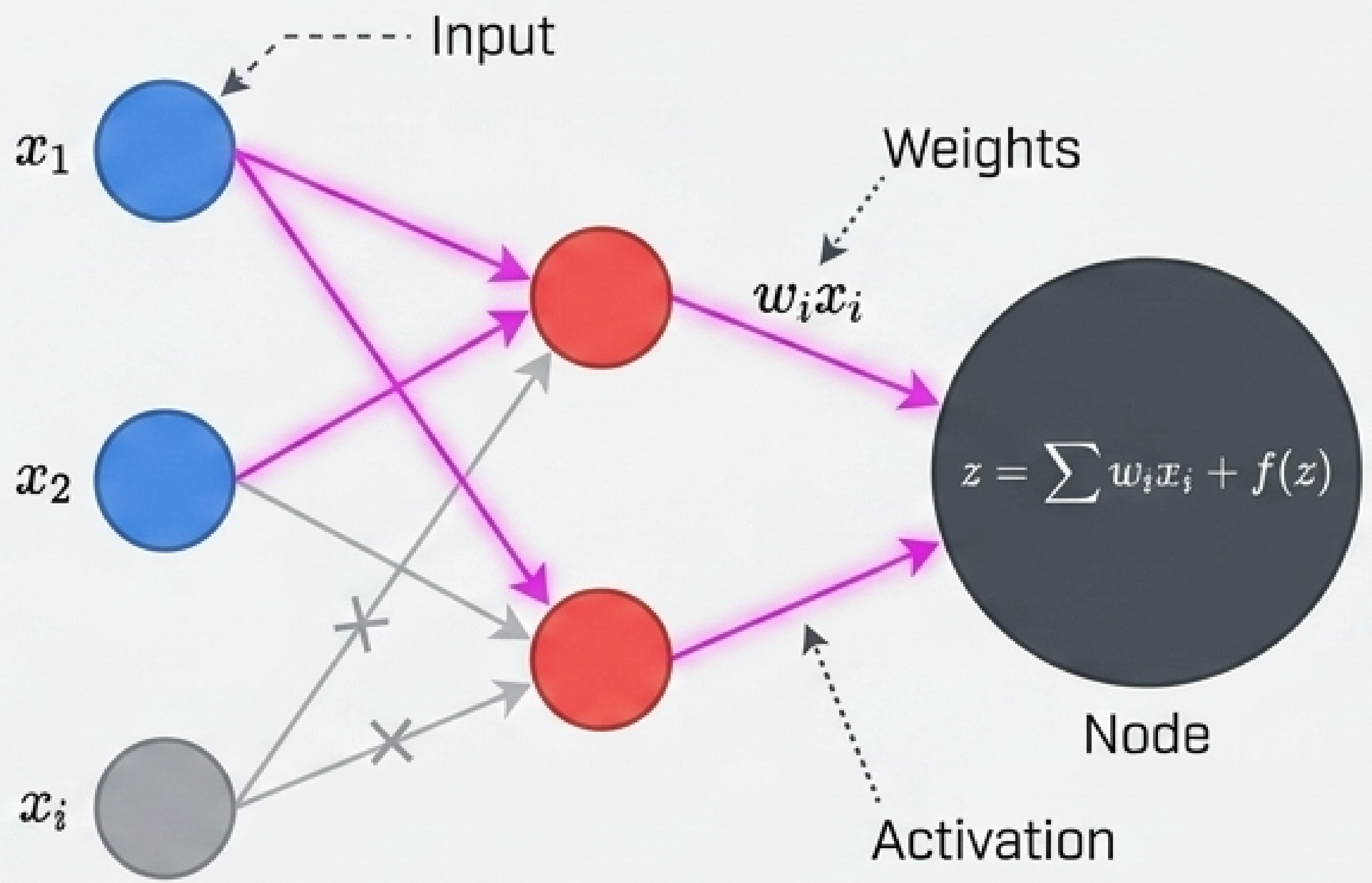
Mathematical Core: Partial Cross-Entropy (pCE)

pCE propagates gradients only from annotated pixels, shielding the model from unclassified background noise.

The Math

$$pfCE = \frac{\sum(Loss_{focal} * Mask_{annotated})}{\sum Mask_{annotated}}$$

The Network



Implementation: PyTorch Loss Formulation

partial_ce_loss.py

partial_ce_loss.py ×

```
class PartialCELoss(nn.Module):
    def __init__(self):
        super(PartialCELoss, self).__init__()

    def forward(self, predictions, targets, mask):
        // Apply boolean mask before mean calculation to bypass zero-labeled background gradients
        focal_loss = self.calculate_focal(predictions, targets)

        masked_loss = focal_loss * mask

        valid_area = torch.sum(mask)

        return torch.sum(masked_loss) / (valid_area + 1e-8)
```

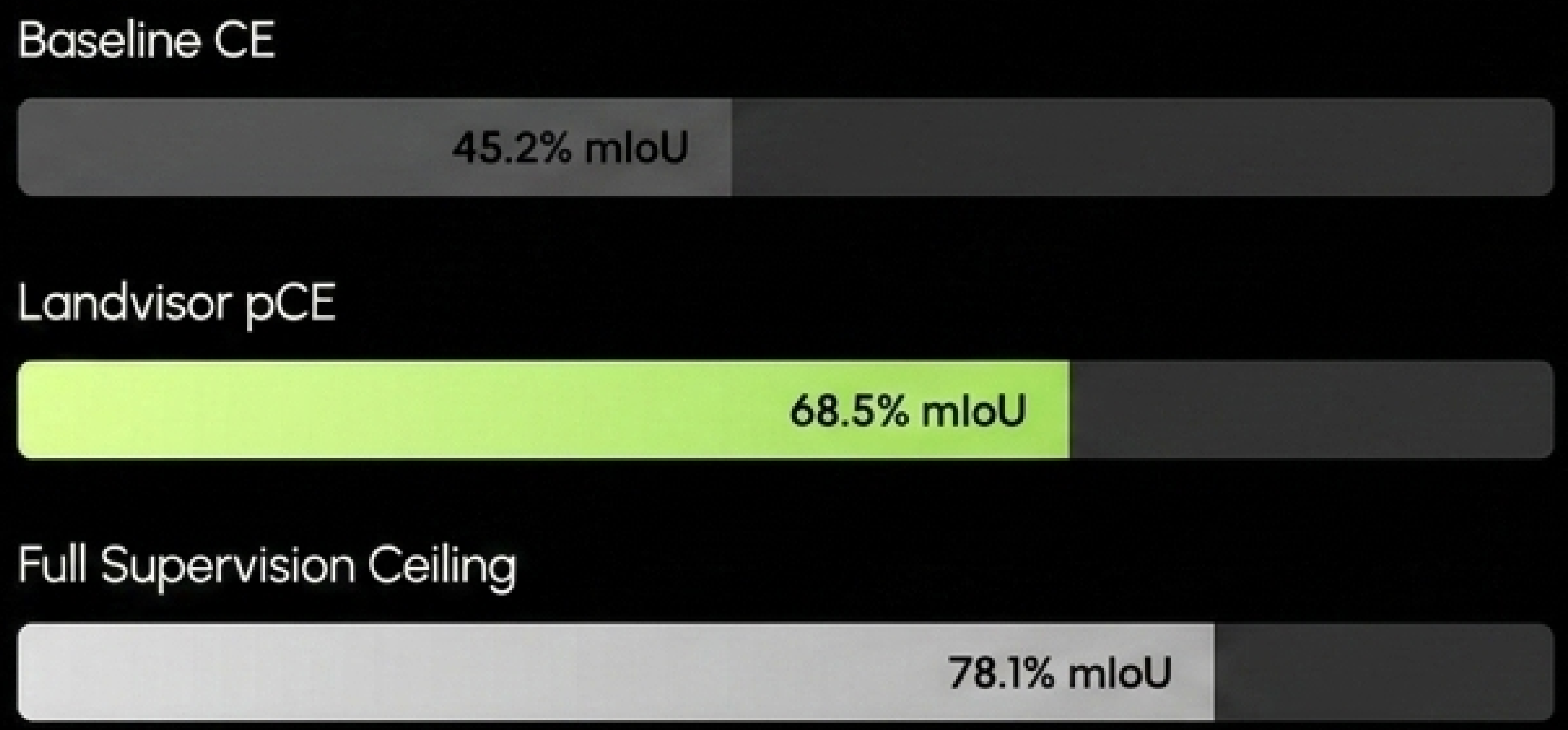
Style Stability via Histogram Matching

Insight: Replaces naive equalization with deterministic CDF mapping to neutralize sun angle and sensor variance without destroying target distributions.

Landvisor Experimental Impact

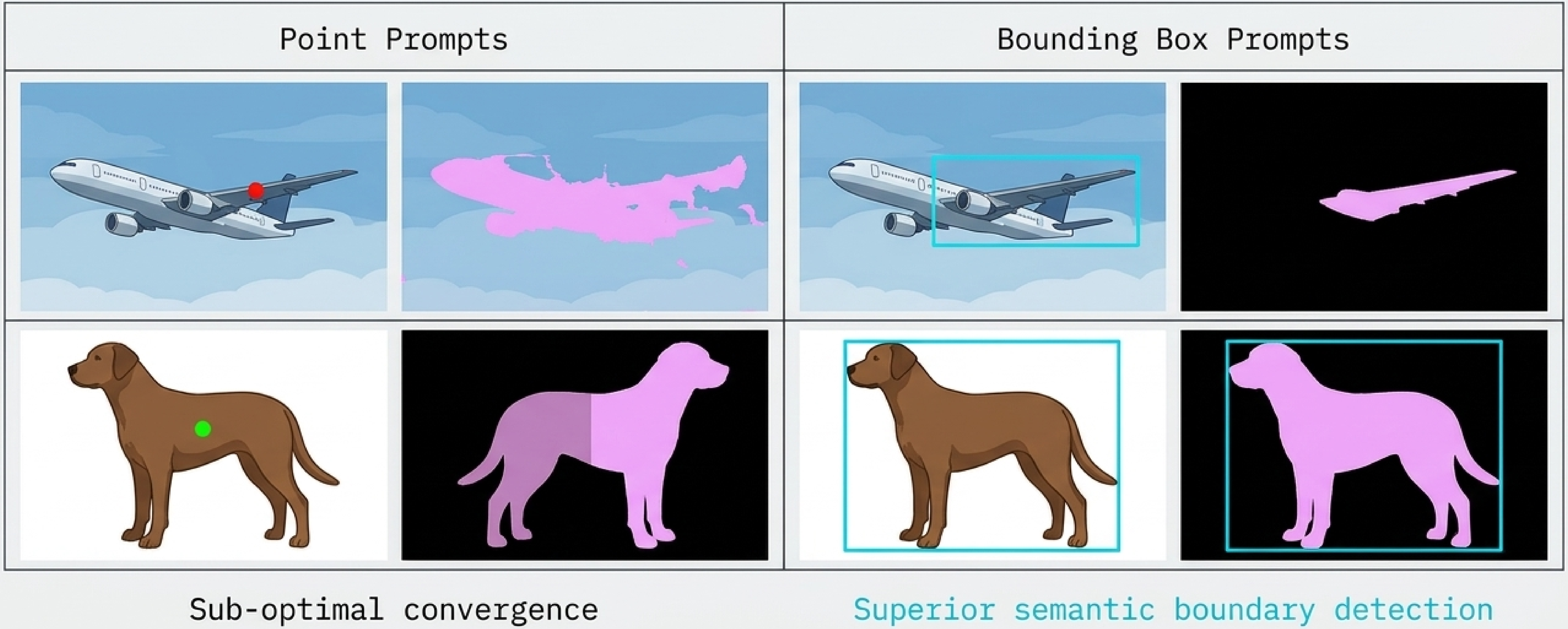
Massive gains in dense urban and complex agricultural regions.
Closes the gap between sparse labels and full supervision by nearly 60%.

+12% mIoU



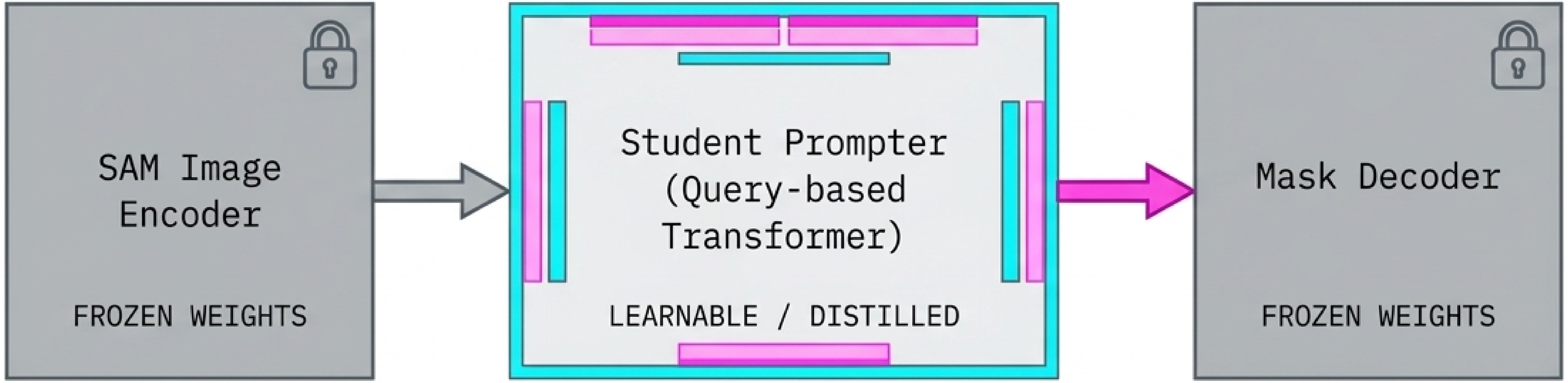
Framework 2: WPS-SAM (Foundation Models)

Insight: Upgrading from points to bounding boxes using zero-shot capabilities to achieve pixel-level part segmentation.



Architecture: Distilling the Student Prompter

Insight: Bypasses flawed external detectors by deriving semantic prompts directly from the frozen image encoder.



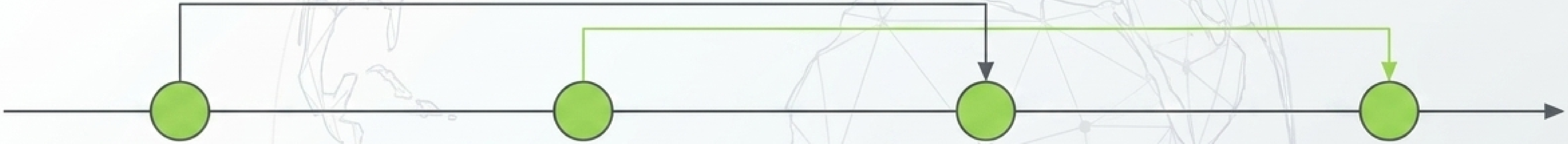
WPS-SAM Performance Matrix

Insight: Outperforms fully supervised methods by ~4% and WSSS by ~12%.

CONSTRAINT: Scaling queries beyond 25 degrades performance due to convergence burden.

Supervision Type	Backbone	mIoU (%)	mAcc (%)
Full	ResNet-50	63.70	72.10
Point	SAM-ViT-B	56.83	64.20
Box	ResNet-101	65.24	74.30
Box	SAM-ViT-B	68.93	79.53

Deployment Roadmap



Phase 1

Loss Function Implementation (pCE) & Unit Testing.

Phase 2

Baseline Training on RS Datasets (DeepGlobe/Vaihingen).

Phase 3

Ensemble Optimization & Test-Time Augmentation (TTA) Tuning.

Phase 4

Full-Scale Integration (Global land use classification).

Architecture Summary: Paradox Resolved

By utilizing Partial CE for point-clouds and distilling Student Prompters for Foundation Models, we achieve near full-supervision accuracy at a fraction of the annotation cost.

[✓] Sparse Data Ingestion Optimized

[✓] Custom Optimization Objectives Deployed

[✓] Foundation Models Distilled

>> PREPARED FOR ARCHITECTURE REVIEW. GIT REPO: /meriti-inc/segmentation-core